

Solution Architecture

Field	Details
Date	15 March 2026
Team ID	PNT2022TMID124356
Project Name	SmartLender AI - Loan Prediction
Submission Stage	3. Project Design Phase

Architecture Layer Outline

- 1. Presentation Layer: Web UI (index.html), AJAX controller (main.js)
- 2. Application Gateway: Flask routing controller endpoints (/predict, /model-telemetry)
- 3. Core Services Layer: LabelEncoder, StandardScaler transforms, classifier models
- 4. Data Layer: joblib artifacts (loan_model.pkl), CSV logs

Component Details

Component	Role	Tech Used
Presentation	Render input forms and re	HTML, CSS, JavaScript
API Gateway	Endpoint controller mappi	Flask Route Handler
Core Logic	Model predict and credit	scikit-learn, Pandas